

UNIT 5 ORDINARY DIFFERENTIAL EQUATIONS

9.1 Modeling with Differential Equations

- **Differential Equation** An equation that contains an **unknown function** and some of its **derivatives**.

- **Exponential Growth Model**

If the rate of growth of population is proportional to the population, then

$$\frac{dP}{dt} = kP$$

for some constant k .

- a reasonable model for a population of bacteria or animals under *ideal* conditions.
- Independent Variable =
- Dependent Variable =
- Show that $P(t) = Ae^{kt}$, for some constant A , is a **solution**.
- Suppose the **initial condition** is $P(0) = 100$.
- **Initial-Value Problem** A problem of finding a solution of the differential equation that satisfies the initial condition.

- **Logistic Model**

The logistic equation is

$$\frac{dP}{dt} = kP \left(1 - \frac{P}{M} \right)$$

for some constants k and M .

- **Equilibrium Solutions**

- **Example** Verify that $y = \sin x \cos x - \cos x$ is a solution of the differential equation $y' + (\tan x)y = \cos^2 x$, on the interval $-\frac{\pi}{2} < x < \frac{\pi}{2}$.

9.3 Separable Equations

- **Example** Solve the differential equation $\frac{dy}{dx} = \frac{y^2}{x^2}$. Then find the particular solution that satisfies the initial condition $y(1) = 2$.

- **Example** Solve the differential equation

$$\frac{dy}{dx} = \frac{6x^2}{2y + \cos y}$$

- **Example** Solve the differential equation

$$\frac{dI}{dt} + 3I = 15$$

Exercises 9.3 1, 3, 5, 7, 11, 13, 15.

9.4 Models for Population Growth

- **Exponential Model**

If $P(t)$ is the value of a quantity at time t and if the rate of change of P with respect to t is proportional to its size $P(t)$, then

$$\frac{dP}{dt} = kP$$

The solution of this equation is

$$P(t) = P_0 e^{kt}$$

where $P_0 = P(0)$ is the initial condition.

- **Logistic Model**

The logistic equation is

$$\frac{dP}{dt} = kP \left(1 - \frac{P}{M} \right)$$

for some constants k and M . The solution of this model is

$$P(t) = \frac{M}{1 + Ae^{-kt}}, \quad A = \frac{M - P_0}{P_0}$$

where $P_0 = P(0)$ is the initial condition.

- **Example** Solve the initial value problem

$$\frac{dP}{dt} = 0.00008P(1000 - P), \quad P(0) = 100$$

and use it to find the population size $P(40)$ and $P(80)$. When does the population reach 900?

Exercises 9.4 3, 5(a)(b)(c).

9.5 Linear Equations

- A first-order differential equation is called **linear** if it can be put into the form

$$\frac{dy}{dx} + P(x)y = Q(x)$$

where P and Q are continuous functions on a given interval.

- **Example** $y' + \frac{1}{x}y = 2$.

- **Example** Solve the differential equation $\frac{dy}{dx} + 3x^2y = 6x^2$.

- **Example** Find the solution of the initial-value problem

$$x^2 y' + xy = 1, \quad x > 0, \quad y(1) = 2$$

- **Example** Solve $y' + 2xy = 1$.

- **Example** Solve

$$\frac{dI}{dt} + 3I = 15 \sin 30t$$

Exercises 9.5 5, 7, 9, 11, 15, 17, 19.

17.1 Second Order Linear Equations

- A **second order** linear differential equation has the form

$$P(x)\frac{d^2y}{dx^2} + Q(x)\frac{dy}{dx} + R(x)y = G(x)$$

- (a) If $G(x) = 0$, the equation is called a **homogeneous** linear equation.
 (b) If $G(x) \neq 0$, the equation is called a **non-homogeneous** linear equation.

Examples of second order linear equations are

$$\begin{aligned}\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + y &= 0 \\ x^2\frac{d^2y}{dx^2} + 3x\frac{dy}{dx} + 2y &= 2x\end{aligned}$$

- **Theorem**

If $y_1(x)$ and $y_2(x)$ are both solutions of the linear homogeneous equation

$$P(x)\frac{d^2y}{dx^2} + Q(x)\frac{dy}{dx} + R(x)y = 0$$

then

$$y(x) = c_1y_1(x) + c_2y_2(x)$$

where c_1 and c_2 are arbitrary constants, is also a solution.

- If $y_1, y_2, \dots, y_n$ are functions,

$$c_1y_1 + c_2y_2 + \cdots + c_ny_n$$

is called a **linear combination** of $y_1, y_2, \dots, y_n$.

- We say that the functions $y_1, y_2, \dots, y_n$ are **linearly independent** if and only if

$$c_1y_1 + c_2y_2 + \cdots + c_ny_n = 0$$

implies that

$$c_1 = c_2 = \cdots = c_n = 0$$

- **Theorem**

If y_1 and y_2 are linearly independent solutions of

$$P(x)\frac{d^2y}{dx^2} + Q(x)\frac{dy}{dx} + R(x)y = 0$$

on an interval I , and $P(x)$ is never 0 on I , then the general solution of the equation is given by

$$y(x) = c_1y_1(x) + c_2y_2(x)$$

where c_1 and c_2 are arbitrary constants.

- **Second Order Linear Homogeneous Equation with Constant Coefficients**

$$a\frac{d^2y}{dx^2} + b\frac{dy}{dx} + cy = 0$$

- **Example** Solve the differential equation

$$y'' + y' - 6y = 0$$

- **Example** Solve the differential equation

$$3\frac{d^2y}{dx^2} + \frac{dy}{dx} - y = 0$$

- **Example** Solve the differential equation

$$4y'' + 12y' + 9y = 0$$

- **Example** Solve the differential equation

$$y'' - 6y' + 13y = 0$$

- **Initial-Value and Boundary-Value Problems**
- **Example** Solve the initial-value problem

$$y'' + y' - 6y = 0, \quad y(0) = 1, \quad y'(0) = 0$$

- **Example** Solve the initial-value problem

$$y'' + y = 0, \quad y(0) = 2, \quad y'(0) = 3$$

- **Example** Solve the boundary-value problem

$$y'' + 2y' + y = 0, \quad y(0) = 1, \quad y(1) = 3$$

Exercises 17.1 1, 3, 5, 7, 9, 11, 13, 17, 19, 21, 23, 25, 27.